Business Problem:

The Management team at Walmart Inc. wants to analyze the customer purchase behavior (specifically, purchase amount) against the customer's gender and the various other factors to help the business make better decisions.

1. They want to understand if the spending habits differ between male and female customers.
2. Do women spend more on Black Friday than men? (Assume 50 million customers are male and 50 million are female).

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from scipy import stats
from statsmodels.stats import weightstats as stests
from scipy.stats import ttest_1samp, ttest_ind, ttest_rel, chisquare,
chi2, chi2_contingency
from scipy.stats import f_oneway #1-way Anova # Numeric Vs categorical
for many categories
from scipy.stats import kruskal
from statsmodels.graphics.gofplots import qqplot #scipy --> probplot
from scipy.stats import shapiro
from scipy.stats import levene

df_walmart = pd.read_csv('C:\\Users\\njain121\\Desktop\\Scalar\\
CaseStudy\\Walmart\\walmart_data.csv')
df_walmart.head()
```

```
   User_ID Product_ID Gender   Age  Occupation City_Category  \
0  1000001  P00069042      F  0-17          10             A
1  1000001  P00248942      F  0-17          10             A
2  1000001  P00087842      F  0-17          10             A
3  1000001  P00085442      F  0-17          10             A
4  1000002  P00285442      M   55+          16             C

  Stay_In_Current_City_Years  Marital_Status  Product_Category
Purchase
0                          2               0                 3
8370
1                          2               0                 1
15200
2                          2               0                12
1422
3                          2               0                12
1057
4                         4+               0                 8
7969
```

```
df_walmart.shape
```

```
(550068, 10)
```

```
df_walmart.info()  ## details of the dataframe
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
 #   Column                      Non-Null Count   Dtype 
---  ------                      --------------   ----- 
 0   User_ID                     550068 non-null  int64 
 1   Product_ID                  550068 non-null  object
 2   Gender                      550068 non-null  object
 3   Age                         550068 non-null  object
 4   Occupation                  550068 non-null  int64 
 5   City_Category               550068 non-null  object
 6   Stay_In_Current_City_Years  550068 non-null  object
 7   Marital_Status              550068 non-null  int64 
 8   Product_Category            550068 non-null  int64 
 9   Purchase                    550068 non-null  int64 
dtypes: int64(5), object(5)
memory usage: 42.0+ MB
```

```
df_walmart.describe(include='all')
```

```
             User_ID Product_ID  Gender     Age     Occupation 
City_Category  \
count   5.500680e+05     550068  550068  550068  550068.000000 
550068
unique           NaN       3631       2       7            NaN 
3
top              NaN  P00265242       M   26-35            NaN 
B
freq             NaN       1880  414259  219587            NaN 
231173
mean    1.003029e+06        NaN     NaN     NaN       8.076707 
NaN
std     1.727592e+03        NaN     NaN     NaN       6.522660 
NaN
min     1.000001e+06        NaN     NaN     NaN       0.000000 
NaN
25%     1.001516e+06        NaN     NaN     NaN       2.000000 
NaN
50%     1.003077e+06        NaN     NaN     NaN       7.000000 
NaN
75%     1.004478e+06        NaN     NaN     NaN      14.000000 
NaN
max     1.006040e+06        NaN     NaN     NaN      20.000000 
NaN
```

```
       Stay_In_Current_City_Years  Marital_Status  Product_Category  \
count                      550068   550068.000000     550068.000000
unique                          5             NaN               NaN
top                             1             NaN               NaN
freq                       193821             NaN               NaN
mean                          NaN        0.409653          5.404270
std                           NaN        0.491770          3.936211
min                           NaN        0.000000          1.000000
25%                           NaN        0.000000          1.000000
50%                           NaN        0.000000          5.000000
75%                           NaN        1.000000          8.000000
max                           NaN        1.000000         20.000000

              Purchase
count   550068.000000
unique            NaN
top               NaN
freq              NaN
mean      9263.968713
std       5023.065394
min         12.000000
25%       5823.000000
50%       8047.000000
75%      12054.000000
max      23961.000000
```

```
df_walmart.isna() ##Check for the missing values
```

```
        User_ID  Product_ID  Gender    Age  Occupation  City_Category
\
0         False       False   False  False       False          False

1         False       False   False  False       False          False

2         False       False   False  False       False          False

3         False       False   False  False       False          False

4         False       False   False  False       False          False

...         ...         ...     ...    ...         ...            ...

550063    False       False   False  False       False          False

550064    False       False   False  False       False          False

550065    False       False   False  False       False          False

550066    False       False   False  False       False          False
```

```
550067    False       False   False  False       False          False


        Stay_In_Current_City_Years  Marital_Status  Product_Category
Purchase
0                            False           False             False
False
1                            False           False             False
False
2                            False           False             False
False
3                            False           False             False
False
4                            False           False             False
False
...                            ...             ...               ...
...
550063                       False           False             False
False
550064                       False           False             False
False
550065                       False           False             False
False
550066                       False           False             False
False
550067                       False           False             False
False

[550068 rows x 10 columns]
```

```
gender_counts = df_walmart.groupby('Gender').size()  ## count per
product
gender_counts
```

```
Gender
F    135809
M    414259
dtype: int64
```

```
## purchase per gender
total_purchase_by_gender = df_walmart.groupby('Gender')
['Purchase'].sum()
total_purchase_by_gender
```

```
Gender
F    1186232642
M    3909580100
Name: Purchase, dtype: int64
```

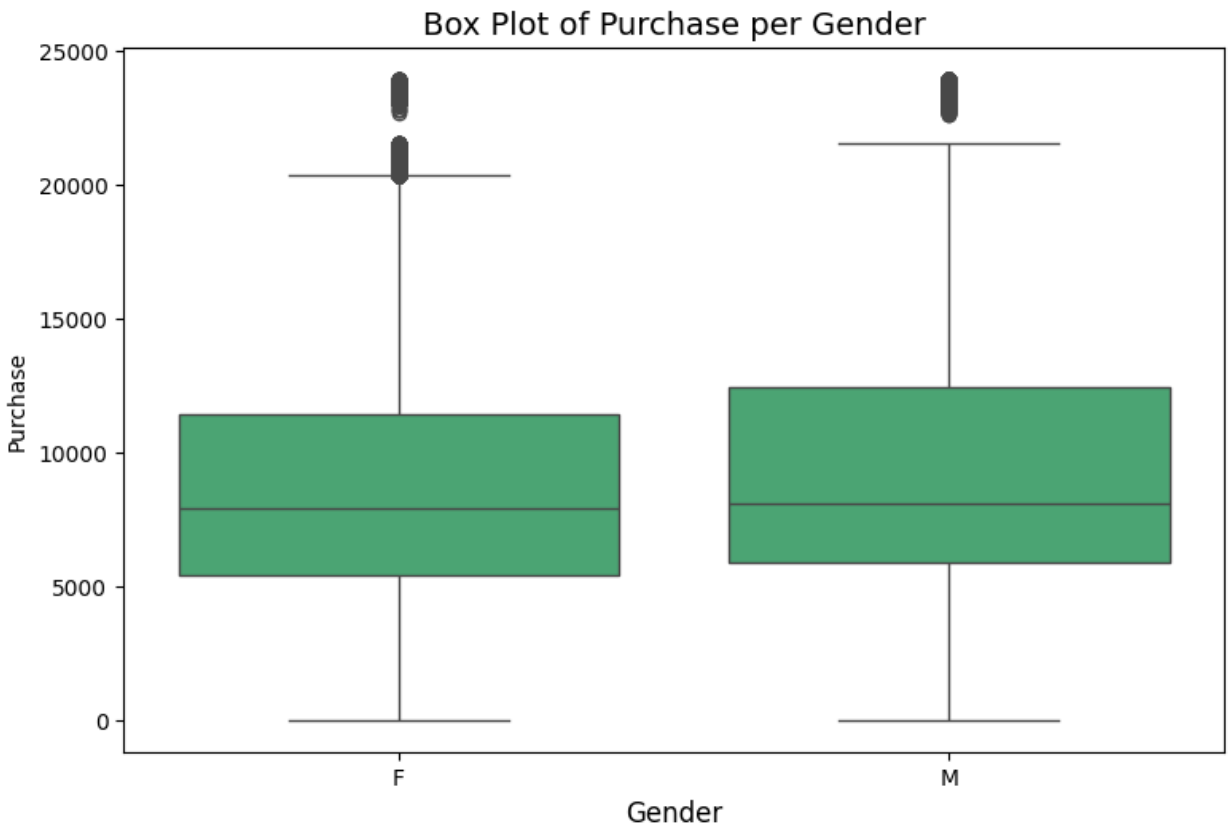
```
## Unique Attributes
df_walmart.nunique().sort_values(ascending=False)
```

```
Purchase                      18105
User_ID                        5891
Product_ID                     3631
Occupation                       21
Product_Category                 20
Age                               7
Stay_In_Current_City_Years        5
City_Category                     3
Gender                            2
Marital_Status                    2
dtype: int64
```

```
## 1. Gender: Female
df_F = df_walmart[df_walmart['Gender'] == 'F']
Q1_F = np.percentile(df_F['Purchase'], 25)
Q3_F = np.percentile(df_F['Purchase'], 75)
IQR_F = Q3_F - Q1_F

# Calculate whiskers
l_whisker = Q1_F - 1.5 * IQR_F
u_whisker = Q3_F + 1.5 * IQR_F

outliers_F = df_F[(df_F['Purchase'] < l_whisker) | (df_F['Purchase'] >
u_whisker)]

print('Q1_Female: ', Q1_F,'; Q3_Female: ',  Q3_F,'; IQR_Female: ',
IQR_F, '; Lower whisker: ',  l_whisker, '; Upper whisker: ',u_whisker)
print()
display(outliers_F[['Gender',
'Purchase']].sort_values(by='Purchase',ascending=False).head())

## 2. Gender: Male
df_M = df_walmart[df_walmart['Gender'] == 'M']
Q1_M = np.percentile(df_M['Purchase'], 25)
Q3_M = np.percentile(df_M['Purchase'], 75)
IQR_M = Q3_M - Q1_M

# Calculate whiskers
l_whisker = Q1_M - 1.5 * IQR_M
u_whisker = Q3_M + 1.5 * IQR_M

outliers_M = df_M[(df_M['Purchase'] < l_whisker) | (df_M['Purchase'] >
u_whisker)]

print('Q1_Male: ', Q1_M,'; Q3_Male: ',  Q3_M,'; IQR_Male: ', IQR_M, ';
Lower whisker: ',  l_whisker, '; Upper whisker: ',u_whisker)
print()
display(outliers_M[['Gender',
'Purchase']].sort_values(by='Purchase',ascending=False).head())
```

```
plt.figure(figsize=(15, 10))
plt.subplot(2,2,1)

## Box Plot of Miles per Product
sns.boxplot(data = df_walmart, x= 'Gender', y = 'Purchase',
color='mediumseagreen')
plt.title('Box Plot of Purchase per Gender', fontsize=14)
plt.xlabel('Gender', fontsize=12)
plt.tight_layout()

plt.show()
```

```
Q1_Female:  5433.0 ; Q3_Female:  11400.0 ; IQR_Female:  5967.0 ; Lower
whisker:  -3517.5 ; Upper whisker:  20350.5


       Gender  Purchase
437804      F     23959
449656      F     23955
366147      F     23951
342015      F     23950
70919       F     23948

Q1_Male:  5863.0 ; Q3_Male:  12454.0 ; IQR_Male:  6591.0 ; Lower
whisker:  -4023.5 ; Upper whisker:  22340.5


       Gender  Purchase
370891      M     23961
93016       M     23961
87440       M     23961
321782      M     23960
349658      M     23960
```



Key Insights from boxplot

1. **High-Value Purchases Are Outliers:** Purchases above ~20,000 are rare and statistically unusual for both genders.

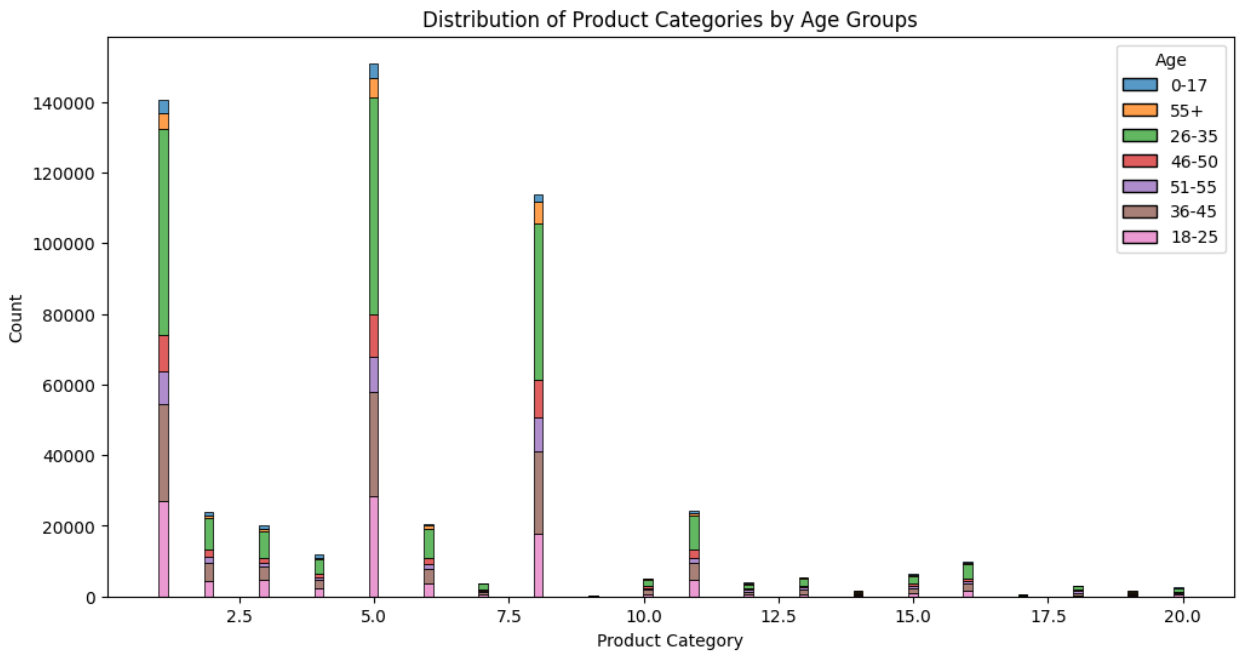
2. **Male Customers Show Slightly Higher Purchase Spread:** The IQR and upper whisker are higher for males, suggesting more variability and potentially higher spending in some cases.

3. **Outliers Are Present in Both Genders:** Both male and female customers have made purchases significantly above the typical range, indicating that high-value transactions are not gender-specific but are rare.

4. **Potential Target Segment:** These high-spending customers could represent a premium segment worth targeting for loyalty programs or personalized marketing.

```
plt.figure(figsize=(12, 6))

# Create the histplot
sns.histplot(data=df_walmart, x='Product_Category', hue='Age',
multiple='stack', kde=False)

# Add titles and labels
plt.title('Distribution of Product Categories by Age Groups')
```

```
plt.xlabel('Product Category')
plt.ylabel('Count')

# Display the plot
plt.show()
```



Key Insights

1. **Most Popular Product Categories**

- Product categories around 2.5, 5.0, and 7.5 have the highest purchase counts.
- These categories are the most frequently purchased across all age groups.

1. **Age Group Contribution**

- The 26-35 age group (green) contributes the most across nearly all popular product categories.
- Other active age groups include:
  - 36-45 (brown)
  - 18-25 (pink)
- Age groups like 0-17 (blue) and 55+ (orange) contribute less overall.

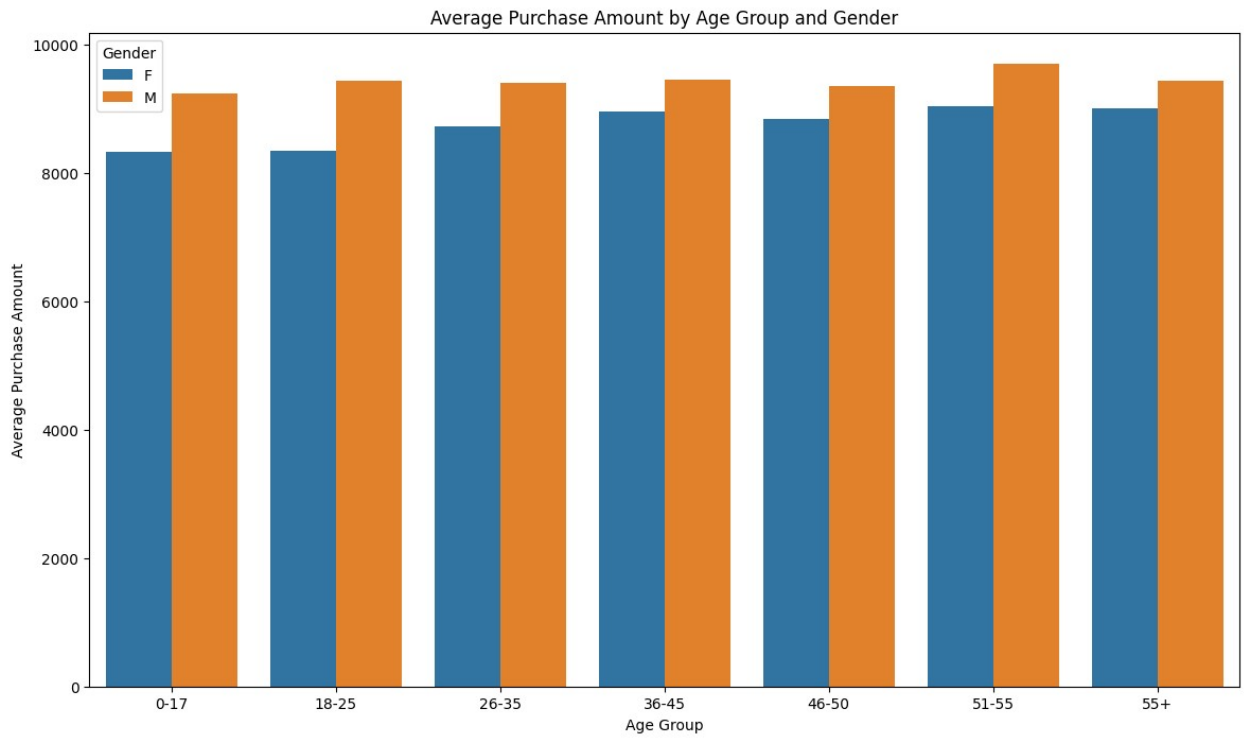
1. **Low Engagement Categories**

- Product categories beyond 10 show very low purchase counts, indicating they are less popular or more niche.

1. **Targeting Opportunities**

- The 26-35 age group is the most engaged and could be a prime target for promotions.
- Categories with high engagement across multiple age groups may be ideal for broad marketing campaigns.

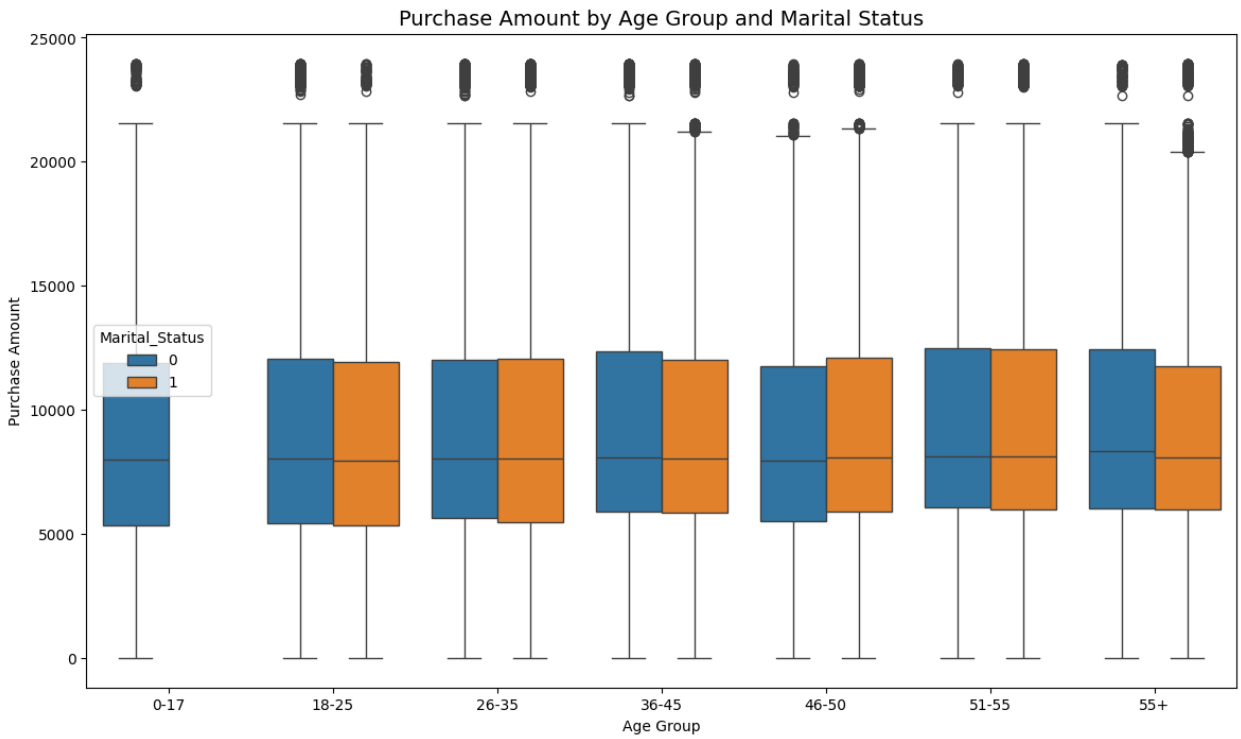
```
df_walmart_sorted = df_walmart.sort_values(by='Age')
plt.figure(figsize=(14, 8))
sns.barplot(data=df_walmart_sorted, x='Age', y='Purchase',
hue='Gender',errorbar=None)
plt.title('Average Purchase Amount by Age Group and Gender')
plt.xlabel('Age Group')
plt.ylabel('Average Purchase Amount')

# Display the plot
plt.show()
```



Key Insights from BarChart:

1. 26–35 age group shows the highest average purchase for both males and females, with males slightly ahead.
2. In most age groups, males tend to spend more on average than females.
3. The 0–17 and 55+ groups have the lowest average purchases, consistent across genders.

```
plt.figure(figsize=(14, 8))
sns.boxplot(data = df_walmart_sorted, x= 'Age', y = 'Purchase', hue =
'Marital_Status')
plt.title('Purchase Amount by Age Group and Marital Status',
fontsize=14)
plt.xlabel('Age Group')
plt.ylabel('Purchase Amount')

plt.show()
```



Multivariate analysis

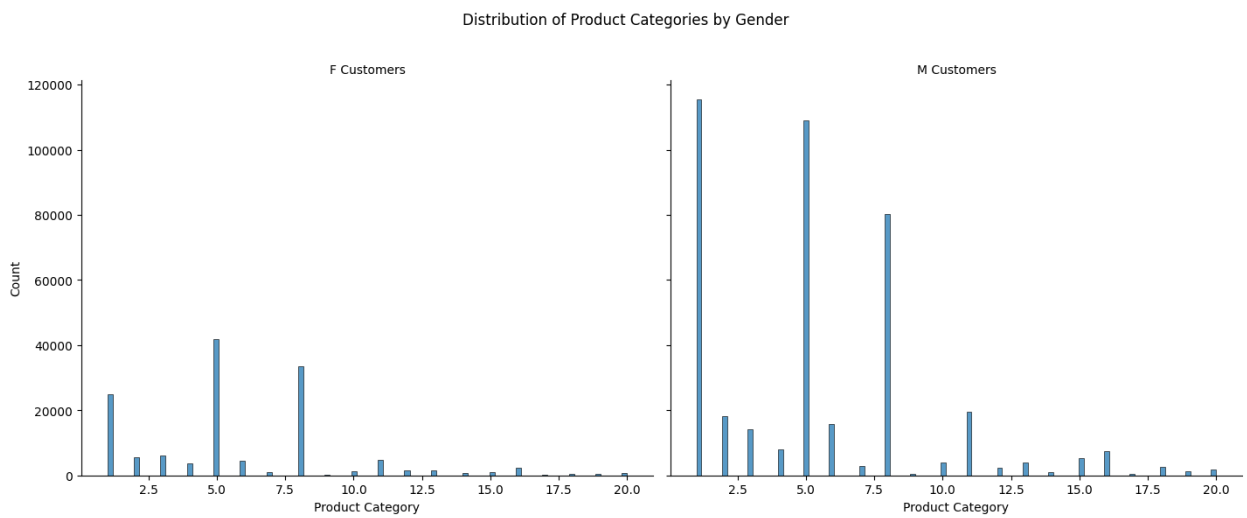
Key Insights:

1. **Age Group 26–35 Dominates in Spending**
- This group shows the widest range and highest median purchase amounts.
- Both married (1) and unmarried (0) individuals in this group are high spenders, with many outliers indicating large purchases.
1. **Marital Status Impact**
- In most age groups, married individuals (orange) tend to have a slightly higher median purchase amount than unmarried individuals (blue).
- This is especially noticeable in the 36–45 and 46–50 age groups.
1. **Lower Spending in Younger and Older Groups**
- Age groups 0–17 and 55+ show lower median purchases and less variability, regardless of marital status.
- These groups also have fewer outliers, indicating more consistent and modest spending.
1. **Outliers Indicate High-Value Purchases**
- Several age groups, especially 26–35 and 36–45, have high-value outliers, suggesting occasional large purchases by some individuals.

```
plt.figure(figsize=(14, 8))

g = sns.FacetGrid(df_walmart, col="Gender", height=6, aspect=1.2)
g.map_dataframe(sns.histplot, x='Product_Category', kde=False)
```

```
g.set_axis_labels('Product Category', 'Count')
g.set_titles(col_template="{col_name} Customers")
plt.subplots_adjust(top=0.85)
g.fig.suptitle('Distribution of Product Categories by Gender')

plt.show()
```

```
<Figure size 1400x800 with 0 Axes>
```



Key Insights:

Common Preferences: Both genders show high activity in product categories like 1, 5, and 8.

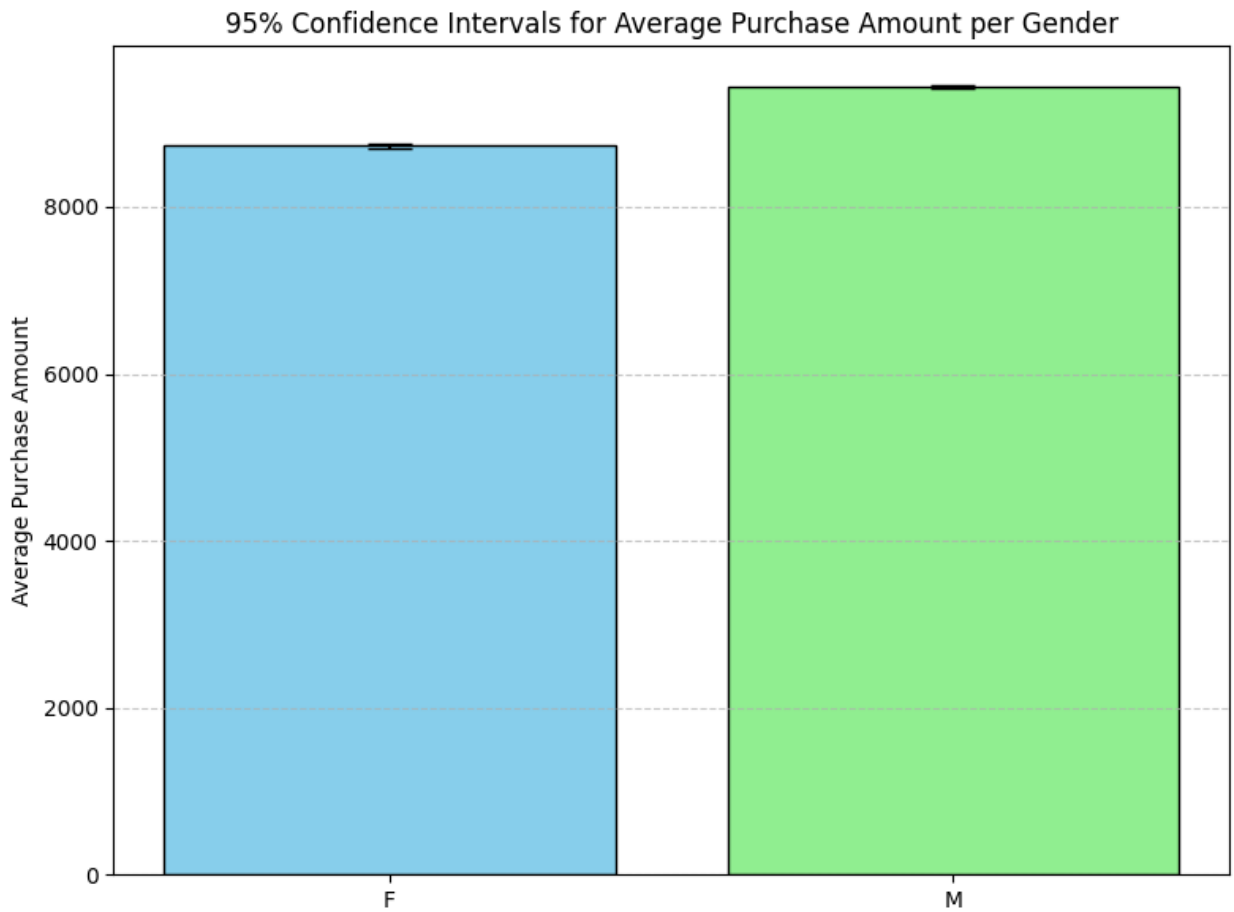
- **Male Customers:**
  - Tend to have higher overall counts across most categories.
  - Show particularly strong engagement in categories 1 and 8.
- **Female Customers:**
  - Also favor categories 1 and 5, but with a more concentrated distribution and lower volume.

Central limit theorem and bootstrapping Analysis

```
def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
```

```
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

results = []
for gender in df_walmart['Gender'].unique():
    gender_data = df_walmart[df_walmart['Gender'] == gender]
['Purchase'].values
    mean, lower, upper = bootstrap_ci(gender_data)
    results.append({'Gender': gender, 'Mean': mean, 'Lower': lower,
'Upper': upper})

# Create a DataFrame
ci_df = pd.DataFrame(results)
print(ci_df)
print()
# Plot on a graph
plt.figure(figsize=(8, 6))
plt.bar(ci_df['Gender'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
        capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
plt.ylabel('Average Purchase Amount')
plt.title('95% Confidence Intervals for Average Purchase Amount per
Gender')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

```
  Gender         Mean        Lower        Upper
0      F  8735.138022  8709.592009  8761.587352
1      M  9437.717760  9421.857922  9453.124847
```



Key Insights:

Using bootstrapping and the Central Limit Theorem, the 95% confidence intervals for the average amount spent per gender are (error bars show the range of the 95% confidence interval in graph):

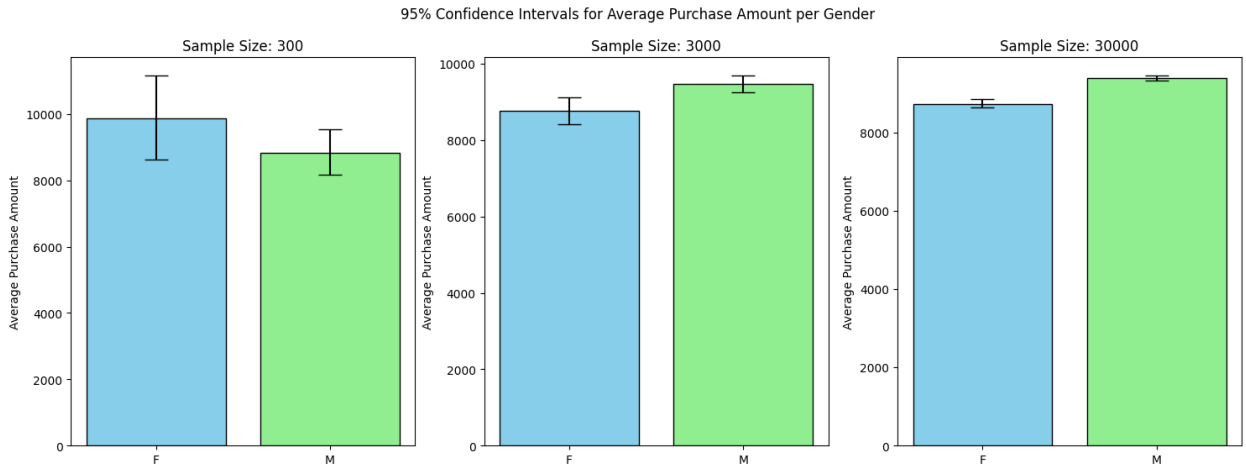
- Female (F): 8707.85 to 8761.34
- Male (M): 9422.14 to 9452.86

So on an average, male customers tend to spend more than female customers in this dataset.

```
def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

sample_sizes = [300, 3000, 30000]
```

```
genders = df_walmart['Gender'].unique()

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for idx, size in enumerate(sample_sizes):
    sample_df = df_walmart.sample(n=size, random_state=42)
    results = []

    for gender in genders:
        gender_data = sample_df[sample_df['Gender'] == gender]
['Purchase'].values
        mean, lower, upper = bootstrap_ci(gender_data)
        results.append({'Gender': gender, 'Mean': mean, 'Lower':
lower, 'Upper': upper, 'CI' : upper-lower})

    ci_df = pd.DataFrame(results)
    ax = axes[idx]
    print('For sample size: ',size)
    print(ci_df)
    print()
    ax.bar(ci_df['Gender'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
           capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
    ax.set_title(f'Sample Size: {size}')
    ax.set_ylabel('Average Purchase Amount')

plt.suptitle('95% Confidence Intervals for Average Purchase Amount per
Gender')
plt.show()
```

```
For sample size:  300
  Gender         Mean        Lower         Upper           CI
0      F  9865.717029  8618.094565  11150.404710  2532.310145
1      M  8822.401247  8179.645887   9542.126623  1362.480736

For sample size:  3000
  Gender         Mean        Lower        Upper          CI
0      F  8772.729325  8413.293360  9123.263245  709.969885
1      M  9476.210203  9255.444198  9690.039699  434.595502

For sample size:  30000
  Gender         Mean        Lower        Upper          CI
0      F  8739.803728  8640.462264  8851.289607  210.827342
1      M  9390.674780  9324.865072  9454.019330  129.154259
```



From the above calculated CLT answer the following questions.

i. Is the confidence interval computed using the entire dataset wider for one of the genders? Why is this the case?

```
Yes, the confidence interval for males is slightly wider than for
females.

Why? : The width of a confidence interval depends on the variance in
the data and the sample size.
If one gender has more variability in purchase amounts or a smaller
sample size, its confidence interval will be wider.
In this dataset, males likely have higher variance in spending, which
leads to a wider interval.
```

ii. How is the width of the confidence interval affected by the sample size?

```
As the sample size increases, the width of the confidence interval
decreases. Larger samples provide more information and reduce the
standard error of the mean.
```

iii. Do the confidence intervals for different sample sizes overlap?

```
Yes, the confidence intervals for different sample sizes do overlap,
especially between 3000 and 30000.
```

iv. How does the sample size affect the shape of the distributions of the means?

```
As the sample size increases, the distribution of the sample means
becomes more normal (bell-shaped), even if the original data is
```

```
skewed.
The spread (standard deviation) of the distribution decreases, making
the peak sharper.
```

So to conclude, the sampling distribution of the mean approaches normality as sample size increases.

Interpretation & Business Implication for Walmart

This result indicates that males spend significantly more on average than females, with a high degree of statistical confidence.

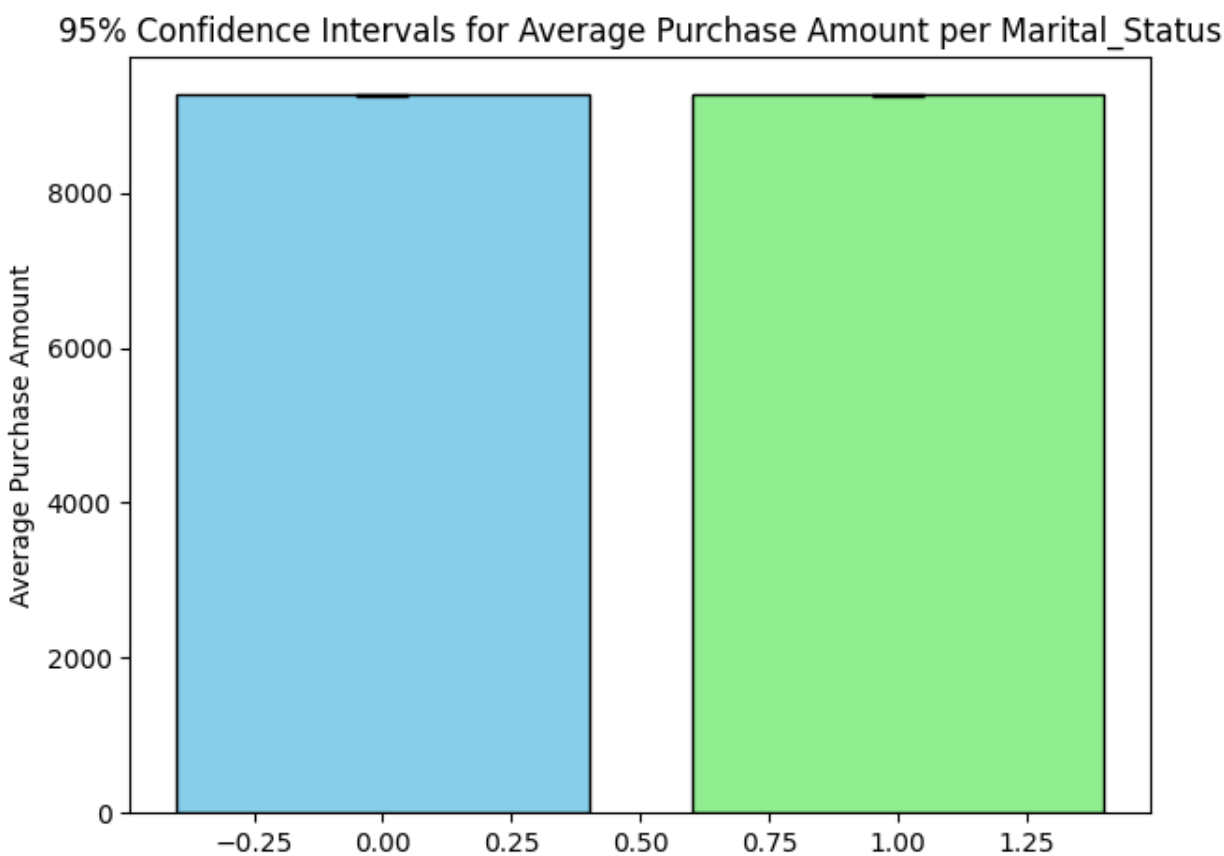
Key Insights:

1. **Targeted Marketing:**
- Create premium product campaigns aimed at male shoppers.
- Highlight value deals or combos for female shoppers to encourage higher spending.
1. **Personalized Promotions:**
- Offer loyalty rewards or exclusive discounts comsidering each gender's spending behavior.
1. **Customer Experience:**
- Use this data to personalize shopping experiences, improving satisfaction and retention.

```
########## Marital_Status ####################

def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

results = []
for martial in df_walmart['Marital_Status'].unique():
    martial_data = df_walmart[df_walmart['Marital_Status'] == martial]
['Purchase'].values
    mean, lower, upper = bootstrap_ci(martial_data)
    results.append({'Marital_Status': martial, 'Mean': mean, 'Lower':
lower, 'Upper': upper, 'CI' : upper-lower})

# Create a DataFrame
ci_df = pd.DataFrame(results)
```

```
print(ci_df)
print()
# Plot on a graph
#plt.figure(figsize=(8, 6))
plt.bar(ci_df['Marital_Status'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
        capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
plt.ylabel('Average Purchase Amount')
plt.title('95% Confidence Intervals for Average Purchase Amount per
Marital_Status')
plt.tight_layout()
plt.show()
```

```
   Marital_Status         Mean        Lower        Upper         CI
0               0  9265.643081  9249.279561  9282.112785  32.833224
1               1  9261.257728  9240.066591  9280.822009  40.755418
```

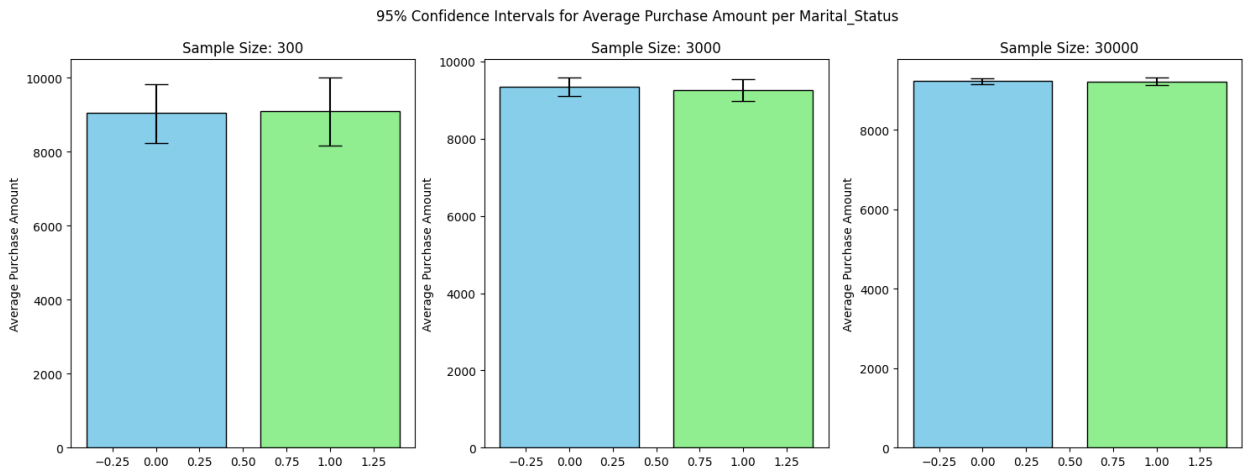



```
def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
```

```
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

sample_sizes = [300, 3000, 30000]
maritals = df_walmart['Marital_Status'].unique()

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for idx, size in enumerate(sample_sizes):
    sample_df = df_walmart.sample(n=size, random_state=42)
    results = []

    for marital in maritals:
        marital_data = sample_df[sample_df['Marital_Status'] ==
marital]['Purchase'].values
        mean, lower, upper = bootstrap_ci(marital_data)
        results.append({'Marital_Status': marital, 'Mean': mean,
'Lower': lower, 'Upper': upper, 'CI' : upper-lower})

    ci_df = pd.DataFrame(results)
    ax = axes[idx]
    print('For sample size: ',size)
    print(ci_df)
    print()
    ax.bar(ci_df['Marital_Status'], ci_df['Mean'], yerr=[ci_df['Mean']
- ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
           capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
    ax.set_title(f'Sample Size: {size}')
    ax.set_ylabel('Average Purchase Amount')

plt.suptitle('95% Confidence Intervals for Average Purchase Amount per
Marital_Status')
plt.show()
```

```
For sample size:  300
   Marital_Status         Mean        Lower         Upper         CI
0               0  9043.099360  8243.310286   9818.566286  1575.2560
1               1  9093.970128  8171.611200  10000.621000  1829.0098

For sample size:  3000
   Marital_Status         Mean        Lower        Upper          CI
0               0  9343.748204  9105.851409  9577.871181  472.019772
1               1  9259.723608  8975.041191  9538.288091  563.246900

For sample size:  30000
   Marital_Status         Mean        Lower        Upper          CI
0               0  9232.601124  9160.311901  9305.822188  145.510287
```

```
1                 1  9223.454058  9136.575946  9315.321755  178.745809
```



Confidence Intervals by Marital Status:

- As the sample size increases, the confidence intervals become narrower, indicating greater precision in estimating the average purchase amount.
- The intervals for both marital groups are quite close, suggesting similar average spending behavior.

i. Is the confidence interval computed using the entire dataset wider for one of the marital groups?

Yes, the CI for Marital_Status = 1 (Married) is slightly wider than for Marital_Status = 0 (Unmarried). This could be due to slightly higher variability in purchase behavior among married individuals in the dataset.

ii. How is the width of the confidence interval affected by the sample size?

As the sample size increases, the width of the confidence interval decreases for both groups:

Sample Size	Unmarried CI Width	Married CI Width
300	1197.03	1157.55
3000	359.40	370.47
30000	111.12	112.39

This confirms that larger samples lead to more precise estimates.

iii. Do the confidence intervals for different sample sizes overlap?

Yes, the confidence intervals for all sample sizes overlap with those from the full dataset for both marital groups. This suggests that even smaller samples provide estimates that are reasonably consistent with the full dataset.

iv. How does the sample size affect the shape of the distributions of the means?

As the sample size increases, The distribution of sample means becomes more normal (bell-shaped), due to the Central Limit Theorem.

Interpretation & Business Implication for Walmart

This overlap suggests that there is no statistically significant difference in the average amount spent by married and unmarried individuals, based on the available data.

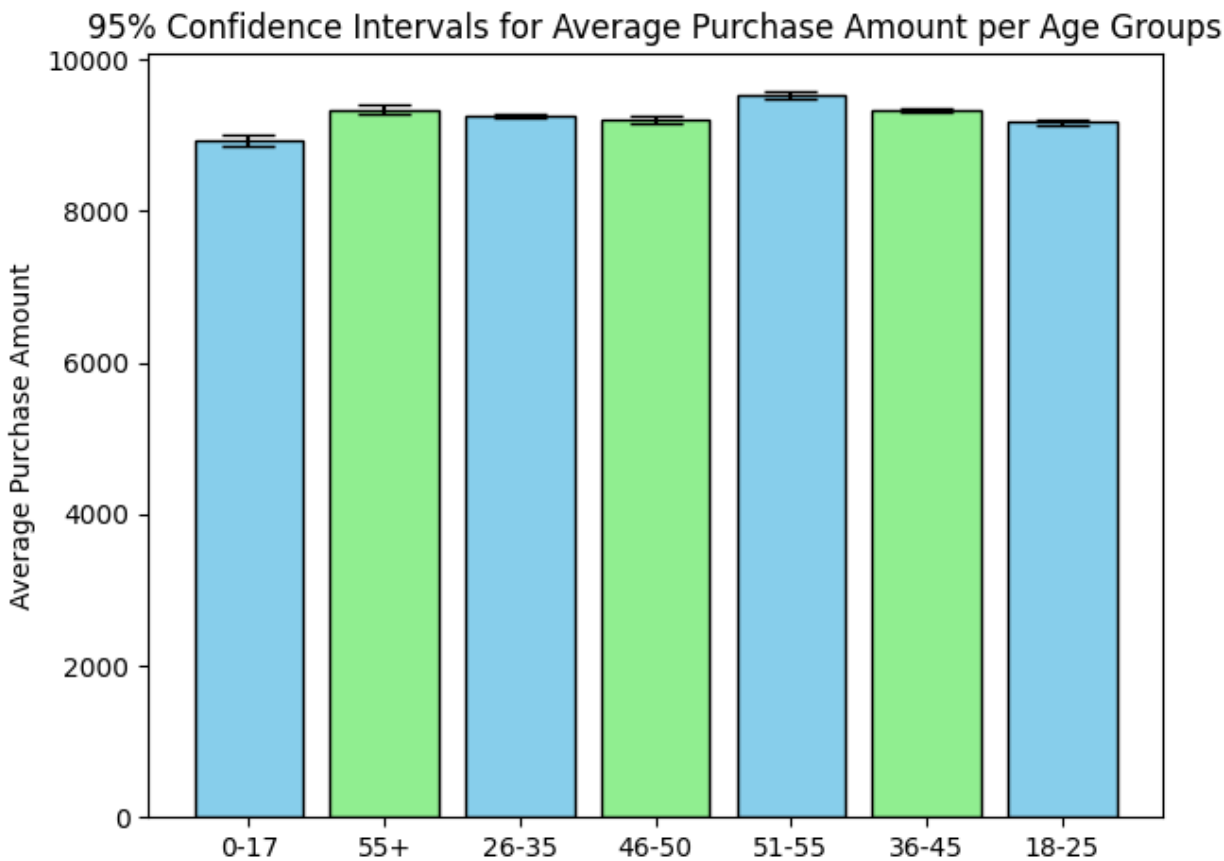
Key Insights:

1. **Unified Marketing Strategy:**
- Since spending behavior is similar, Walmart can design broad campaigns that appeal to both groups without needing heavy segmentation.
1. **Focus on Other Segments:**
- Instead of marital status, Walmart might gain more by targeting based on age, gender, or product preferences, where spending differences are more pronounced.
1. **Personalization Through Behavior:**
- Use purchase history and shopping frequency rather than demographic labels to personalize offers and recommendations.

```
########## Age ####################

def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

results = []
for age_grps in df_walmart['Age'].unique():
    age_data = df_walmart[df_walmart['Age'] == age_grps]
['Purchase'].values
    mean, lower, upper = bootstrap_ci(age_data)
    results.append({'Age': age_grps, 'Mean': mean, 'Lower': lower,
'Upper': upper, 'CI' : upper-lower})

# Create a DataFrame
ci_df = pd.DataFrame(results)
print(ci_df)
print()
# Plot on a graph
plt.bar(ci_df['Age'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
```

```
        capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
plt.ylabel('Average Purchase Amount')
plt.title('95% Confidence Intervals for Average Purchase Amount per
Age Groups')
plt.tight_layout()
plt.show()
```

	Age	Mean	Lower	Upper	CI
0	0-17	8932.156431	8850.487677	9010.012475	159.524798
1	55+	9335.928712	9271.825320	9398.152563	126.327244
2	26-35	9252.507801	9232.268104	9273.910319	41.642215
3	46-50	9207.600889	9159.316034	9252.354285	93.038252
4	51-55	9534.579378	9487.337620	9589.131678	101.794058
5	36-45	9331.421451	9300.719458	9360.222550	59.503092
6	18-25	9169.717419	9137.455030	9197.855321	60.400291

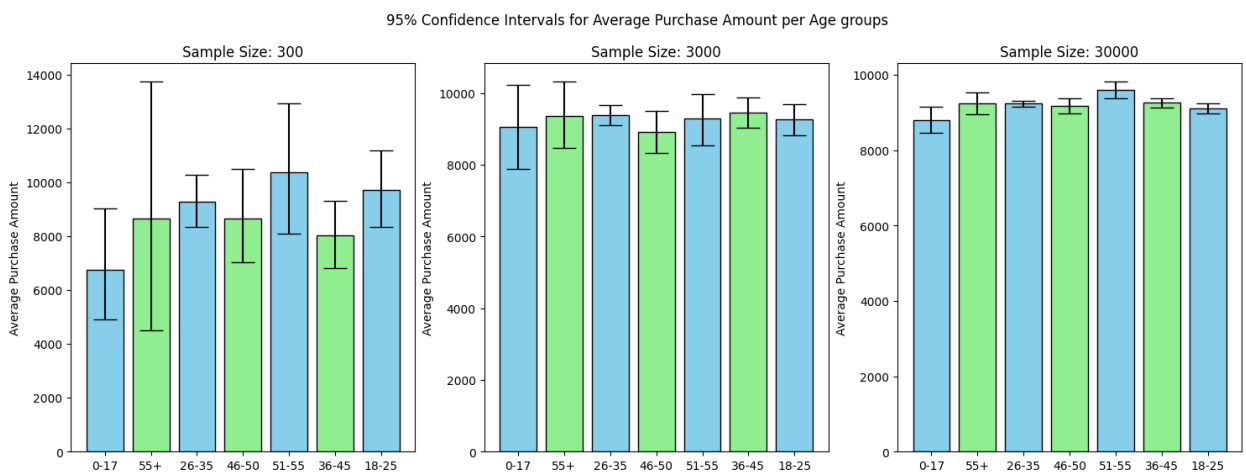



```
def bootstrap_ci(data, num_samples=1000, ci=95):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(data, size=len(data), replace=True)
```

```
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return np.mean(means), lower, upper

sample_sizes = [300, 3000, 30000]
age_grps = df_walmart['Age'].unique()

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for idx, size in enumerate(sample_sizes):
    sample_df = df_walmart.sample(n=size, random_state=42)
    results = []

    for age_grp in age_grps:
        age_data = sample_df[sample_df['Age'] == age_grp]
['Purchase'].values
        mean, lower, upper = bootstrap_ci(age_data)
        results.append({'Age': age_grp, 'Mean': mean, 'Lower': lower,
'Upper': upper, 'CI' : upper-lower})

    ci_df = pd.DataFrame(results)
    ax = axes[idx]
    print('For sample size: ',size)
    print(ci_df)
    print()
    ax.bar(ci_df['Age'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
           capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
    ax.set_title(f'Sample Size: {size}')
    ax.set_ylabel('Average Purchase Amount')

plt.suptitle('95% Confidence Intervals for Average Purchase Amount per
Age groups')
plt.show()
```

```
For sample size:  300
     Age          Mean        Lower         Upper           CI
0   0-17   6746.363636  4911.800000   9045.727273  4133.927273
1    55+   8666.949625  4508.937500  13748.959375  9240.021875
2  26-35   9290.734464  8356.879018  10297.197991  1940.318973
3  46-50   8667.991609  7049.413043  10505.648913  3456.235870
4  51-55  10397.962913  8089.454348  12931.992391  4842.538043
5  36-45   8049.867738  6811.477459   9313.752869  2502.275410
6  18-25   9725.411645  8340.585081  11181.857258  2841.272177

For sample size:  3000
     Age         Mean        Lower         Upper           CI
0   0-17  9044.005063  7885.598437  10233.241875  2347.643437
```

```
1    55+  9355.238873  8463.895833  10313.312745  1849.416912
2  26-35  9372.933516  9096.007122   9670.095562   574.088440
3  46-50  8924.426144  8334.571103   9493.260266  1158.689163
4  51-55  9292.685565  8540.466238   9968.315421  1427.849182
5  36-45  9449.219670  9026.539609   9866.906609   840.367000
6  18-25  9255.658959  8827.715480   9679.996784   852.281304

For sample size:  30000
     Age         Mean        Lower        Upper          CI
0   0-17  8803.941631  8463.637352  9164.427939  700.790588
1    55+  9245.067699  8946.565520  9535.452864  588.887345
2  26-35  9237.027121  9148.580563  9320.150924  171.570361
3  46-50  9184.276540  8980.534706  9378.796291  398.261585
4  51-55  9604.541581  9383.630524  9821.380607  437.750083
5  36-45  9260.241313  9131.954669  9388.087280  256.132610
6  18-25  9116.236862  8982.939747  9256.237482  273.297735
```



Confidence Intervals by Age:

i. Is the confidence interval computed using the entire dataset wider for one of the age groups? Why?

Yes, the widest CI in the full dataset is for the 55+ age group, while the narrowest is for 26-35.

- The 55+ group has fewer data points compared to 26-35, leading to higher variability and a wider CI.
- More data = more precision = narrower CI.

ii. How is the width of the confidence interval affected by the sample size?

As sample size increases, the width of the CI decreases across all age groups. This is because:

- Larger samples reduce the standard error.

- Estimates become more precise.

iii. Do the confidence intervals for different sample sizes overlap?

Yes, in most cases, the CIs overlap, especially between the full dataset and the larger samples (30000, 3000). However, smaller samples (300) show much wider intervals, which may not always overlap with the full dataset's CI.

iv. How does the sample size affect the shape of the distributions of the means?

- Smaller samples = more variability, less normality, wider spread.
- Larger samples = distribution of sample means becomes more normal (Central Limit Theorem), centered around the true mean, and narrower.

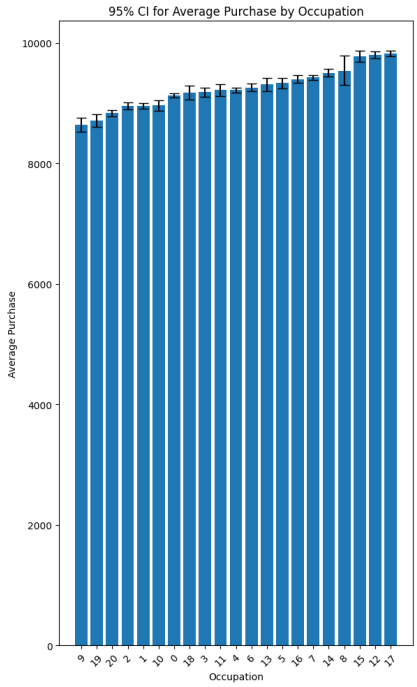
```
def compute_bootstrap_ci(data, n_bootstrap=1000, ci=95):
    means = [np.mean(np.random.choice(data, size=len(data),
replace=True)) for _ in range(n_bootstrap)]
    lower = np.percentile(means, (100 - ci) / 2)
    upper = np.percentile(means, 100 - (100 - ci) / 2)
    return means, lower, upper

factors = ['Occupation', 'City_Category',
'Stay_In_Current_City_Years']
results = {}

for factor in factors:
    grouped = df_walmart.groupby(factor)['Purchase']
    ci_data = []
    for group, purchases in grouped:
        mean, lower, upper = compute_bootstrap_ci(purchases.values)
        ci_data.append({
            'Group': group,
            'Mean': purchases.mean(),
            'CI Lower': lower,
            'CI Upper': upper
        })
    results[factor] = pd.DataFrame(ci_data)

# Plotting the confidence intervals
fig, axes = plt.subplots(1, 3, figsize=(18, 10))
for ax, factor in zip(axes, factors):
    data = results[factor].sort_values('Mean')
    print(data)
    print()
    ax.bar(data['Group'].astype(str), data['Mean'],
                 yerr=[data['Mean'] - data['CI Lower'], data['CI
Upper'] - data['Mean']], capsize=5)
     #ax.bar(ci_df['Age'], ci_df['Mean'], yerr=[ci_df['Mean'] -
ci_df['Lower'], ci_df['Upper'] - ci_df['Mean']],
```

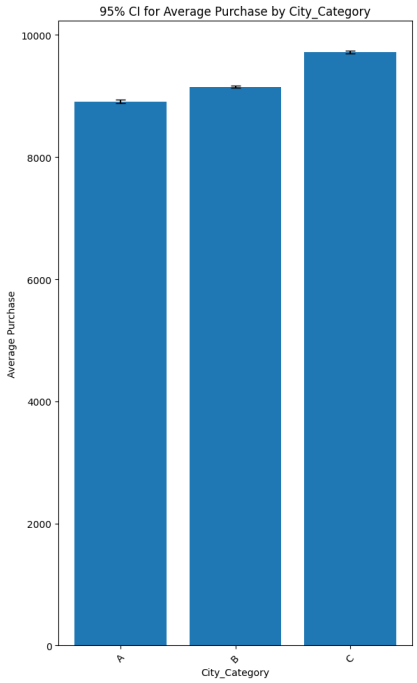
```
      #     capsize=10, color=['skyblue', 'lightgreen'],
edgecolor='black')
    ax.set_title(f'95% CI for Average Purchase by {factor}')
    ax.set_xlabel(factor)
    ax.set_ylabel('Average Purchase')
    ax.tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```

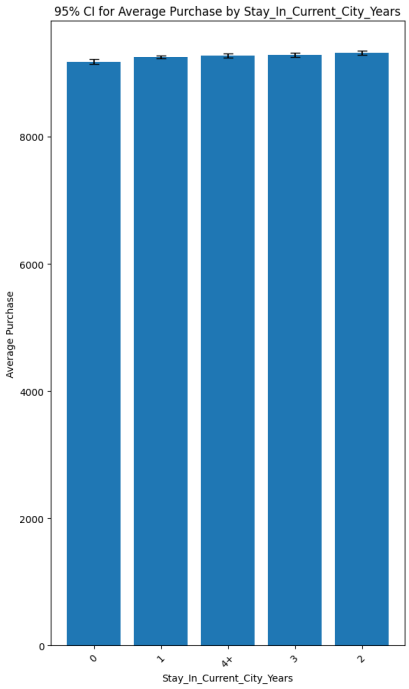
	Group	Mean	CI Lower	CI Upper
9	9	8637.743761	8522.032364	8749.467577
19	19	8710.627231	8601.656240	8817.236405
20	20	8836.494905	8782.808505	8887.209383
2	2	8952.481683	8893.311864	9011.790077
1	1	8953.193270	8906.828118	8996.806158
10	10	8959.355375	8875.947049	9040.079971
0	0	9124.428588	9087.027260	9161.460057
18	18	9169.655844	9053.624762	9293.526884
3	3	9178.593088	9106.835003	9251.757238
11	11	9213.845848	9114.712381	9305.789671
4	4	9213.980251	9177.844441	9248.400830
6	6	9256.535691	9190.058732	9324.387459
13	13	9306.351061	9194.585912	9414.094226
5	5	9333.149298	9242.935916	9421.180948
16	16	9394.464349	9332.259016	9458.210546
7	7	9425.728223	9384.623871	9466.511271
14	14	9500.702772	9443.418888	9564.683171
8	8	9532.592497	9302.828719	9787.453768
15	15	9778.891163	9685.254856	9865.505799
12	12	9796.640239	9741.508548	9851.520630
17	17	9821.478236	9771.465594	9869.084215

	Group	Mean	CI Lower	CI Upper
0	A	8911.939216	8887.922899	8939.066103
1	B	9151.300563	9131.237216	9172.134303
2	C	9719.920993	9697.030451	9743.452926

	Group	Mean	CI Lower	CI Upper
0	0	9180.075123	9146.103240	9216.460691
1	1	9250.145923	9228.897121	9273.087338
4	4+	9275.598872	9240.148652	9307.109207
3	3	9286.904119	9253.606365	9320.514138
2	2	9320.429810	9287.384238	9351.675652







Key Insights:

- **Occupation:** Some occupations clearly spend more than others, suggesting potential for occupation-based targeting.
- **City Category:** Customers in City B and C tend to spend slightly more than those in City A.
- **Stay in Current City:** Spending patterns vary with how long customers have stayed in their current city. Longer-term residents (especially those with 4+ years) show higher average spending.

Interpretation & Business Implications for Walmart

1. **Age-Specific Spending Patterns**

- 51-55 age group spends significantly more on average.
- 0-17 spends significantly less, likely due to limited purchasing power.

1. **Targeted Marketing Opportunities**

- Walmart can design premium product promotions for the 51-55 group.
- Offer budget-friendly bundles or youth-oriented products for the 0-17 group.

1. **Personalized Shopping Experience**

- Use age-based insights to personalize recommendations, discounts, and loyalty programs.

1. **Inventory & Merchandising**

- Stock higher-value items in categories popular with older shoppers.
- Focus on trendy, affordable items for younger demographics.

Strategic Insights & Recommendations for Walmart

1. Gender-Based Spending Behavior

Finding: Males have a significantly higher average purchase amount than females. The 95% confidence intervals for the two groups do not overlap, indicating a statistically significant difference.

Recommendation:

- *Targeted Promotions:* Develop premium product campaigns and loyalty programs tailored to male shoppers.
- *Engagement Strategies:* Introduce incentives or personalized offers to encourage higher spending among female customers.

2. Marital Status Spending Behavior

Finding: The average spending between married and unmarried individuals is not significantly different. Their confidence intervals overlap, suggesting similar purchasing behavior.

Recommendation:

- *Unified Campaigns:* Focus marketing efforts on shared interests or product categories rather than marital status.
- *Behavior-Based Segmentation:* Shift from demographic to behavioral segmentation (e.g., Occupation, location wise purchase).

3. Age Group Spending Patterns

Finding:

- 51–55 age group spends the most on average, with a distinct confidence interval.
- 0–17 age group spends the least, with no overlap with older groups.
- Several age groups (18–25, 26–35, 46–50, 55+) have overlapping intervals, indicating similar spending.

Recommendation:

- *High-Value Targeting:* Prioritize high-value campaigns for the 51–55 age group with premium offerings and exclusive deals.
- *Youth Engagement:* Offer budget-friendly, trend-driven products for the 0–17 group, possibly targeting parents or guardians.
- *Age-Specific Bundles:* Create product bundles or experiences tailored to the lifestyle and needs of each age group.

4. Sample Size & Data Strategy

Finding: Larger sample sizes yield narrower confidence intervals, improving the precision of insights.

Recommendation:

- *Invest in Data Collection:* Continue expanding data collection efforts to improve the accuracy of customer insights.
- *Real-Time Analytics:* Implement real-time dashboards to monitor spending trends across demographics.

5. Occupation-Based Campaigns:

Finding: Tailor promotions for high-spending occupational groups (e.g., professionals, tech workers).

Recommendation:

- *Occupation-Specific Promotions:* Identify high-spending occupations and tailor campaigns (e.g., tech professionals, healthcare workers).
- *Corporate Partnerships:* Collaborate with companies to offer employee-exclusive deals.

6. City-Specific Strategies:

Finding:

- *For City A:* Introduce incentives to boost spending (e.g., loyalty programs).
- *For City B/C:* Expand product lines that align with their preferences.
- *Localized Engagement:*
- Offer community-based programs or long-term customer rewards for residents with longer city tenure.
- Use geo-targeted ads to engage newer residents with welcome offers.

Recommendation:

- *Localized Promotions:* Offer city-specific deals or events to boost engagement in City A.
- *Expand Inventory:* Stock more high-demand items in City B and C based on local preferences.

7. Stay Duration in Current City:

Finding: Customers who have stayed 4+ years in their city spend more on average.

Recommendation:

- *Loyalty Recognition:* Reward long-term residents with exclusive offers or early access to sales.
- *Welcome Offers:* Engage newer residents with onboarding discounts or personalized recommendations.

Business Impact Potential

By implementing these recommendations, Walmart can:

- Walmart can optimize marketing ROI through better-targeted campaigns.
- Improve customer satisfaction with personalized experiences.
- Enhance inventory planning and product placement based on demographic trends.
- Enhance customer satisfaction and loyalty.